

BLAKKBOX HYBRID DENSO DIESEL CALIBRATION & FIRMWARE ANALYSIS WORKFLOW

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SUPPORTED PLATFORMS

- TOYOTA KD/GD
- MITSUBISHI 4D56 / 4M41 / 4N15
- NISSAN YD25 / ZD30

SUPPORTED MCU ARCHITECTURES

- SH7055
- SH7058
- SH7059
- SH72531
- SH72543
- SH72546
- MPC5xx
- RH850
- TriCore

SUPPORTED BIN SIZES

- 368KB
- 512KB
- 716KB
- 1MB
- 1.5MB
- 2MB
- 2.5MB
- 4MB

CORE CALIBRATION PHILOSOPHY

FINAL = ORIGINAL + FILTERED_DELTA

PRIMARY OBJECTIVES

- Preserve OEM structure
- Preserve untouched regions byte-for-byte
- Preserve interpolation continuity
- Reduce smoke
- Reduce fuel knock
- Reduce rattling
- Preserve safe torque behavior
- Maintain combustion stability
- Maintain checksum-safe structure

CORE OPERATING PRINCIPLE

1. Prove layout first
2. Prove structure second
3. Prove calibration meaning last

BASE EXECUTION RULES

- ORIGINAL.bin = READ ONLY
- MOD.bin = ACTIVE WORKSPACE

- NEVER overwrite ORIGINAL
- NO checksum correction during processing
- Preserve all OEM untouched areas
- Preserve zero-groups
- Preserve binary logic regions
- Preserve switch structures
- Preserve DTC masks
- Preserve EGR regions

FIRMWARE NORMALIZATION STAGE

INPUT SUPPORT:

- BIN
- HEX
- S19
- MOT
- ELF

NORMALIZATION:

- Convert all formats into flat binary
- Preserve address continuity
- Preserve byte alignment
- Verify stable byte count
- Reject truncated exports
- Reject offset-shifted binaries

MANDATORY PRE-CHECKS

VERIFY:

- SW ID
- ROM size
- Address alignment
- No offset drift
- Matching ECU family
- Matching transmission type
- Matching platform
- Stable geometry layout

REJECT IF:

- ROM size mismatch
- Corrupted structure
- Unknown displacement
- Invalid geometry
- Unstable modified regions

ARCHITECTURE FINGERPRINTING

Detect using:

- reset vectors
- interrupt layout
- opcode signatures
- pointer spacing
- known firmware signatures

ENDIANNESS VALIDATION

Validate:

- pointer plausibility
- structure alignment
- checksum consistency
- stable instruction decoding

MEMORY SEGMENTATION

Classify:

- bootloader
- application
- calibration
- NVM
- protected regions

ENTROPY CLASSIFICATION

Low entropy:

- structured calibration

Medium entropy:

- executable code

High entropy:

- compressed/encrypted/protected regions

ROM INTEGRITY ENGINE

- SHA256 verification
- alignment verification
- executable protection
- region continuity validation
- geometry verification
- offset stability verification

PROTECTED REGION LOGIC

IF MOD BYTE == 0x00:

DO NOT MODIFY

Possible protected regions:

- EGR structures
- DTC masks
- switch logic
- limiter structures
- checksum structures

SMART DELTA EXTRACTION

$\Delta = \text{MOD} - \text{ORIGINAL}$

Only modified regions are processed.

Untouched OEM regions remain byte-identical.

FALSE-POSITIVE SUPPRESSION

Reject:

- random byte clusters
- invalid surfaces
- broken geometry
- fragmented regions
- executable-like regions

AXIS VALIDATION

Validate:

- monotonic increase
- engineering plausibility
- logical spacing
- scaling consistency

MAP DETECTION LOGIC

Detect:

- smooth interpolation
- gradient continuity
- neighbor consistency
- structured variance
- grouped surfaces

DIMENSION VALIDATION

Verify:

- row count
- column count
- alignment consistency
- table spacing continuity

AGGRESSIVE DELTA ENGINE

$\Delta 0-5$:

$\times 1.15-1.25$

$\Delta 5-10$:

$\times 1.22-1.35$

$\Delta > 10$:

$\times 1.30-1.42$

Additional spike restraint above $\Delta 40$.

Purpose:

- stronger torque response
- preserve combustion stability
- controlled aggressive behavior

SPORTSTUNE FILTER LOGIC

$\Delta 0-5$:

Preserve

$\Delta 5-8$:

$\times 0.80$ attenuation

$\Delta > 8$:

$\times 0.55$ attenuation

Purpose:

- reduce smoke spikes
- preserve drivability
- preserve interpolation smoothness

ECONOMY TUNE LOGIC

$\Delta 0-5$:

$\times 0.88-0.92$

$\Delta 5-10$:

$\times 0.74-0.82$

$\Delta > 10$:

$\times 0.62-0.72$

Purpose:

- improve combustion efficiency
- reduce overfueling
- reduce smoke
- improve low-load smoothness

MAP GEOMETRY RECONSTRUCTION

- grouped surface detection
- interpolation continuity analysis
- axis stability validation
- neighboring trend preservation

INTERPOLATION PRESERVATION

DO:

- smooth jagged transitions
- preserve gradient continuity
- preserve neighboring trend behavior

DO NOT:

- create spikes
- create cliffs
- create sharp transitions
- break axis progression

COMBUSTION REFINEMENT STRATEGY

TARGET AREAS:

- startup
- idle
- low load
- 1000–2500 RPM
- shifting
- WOT transition

OBJECTIVES:

- reduce black smoke
- reduce white smoke
- reduce fuel knock
- reduce diesel clatter
- stabilize transient combustion

LOW LOAD REFINEMENT

RPM:

900–2200

LOAD:

0–50%

TARGET:

- smoother torque ramp
- cleaner combustion
- lower NVH
- stable rail behavior
- lower soot output

AIRFLOW SYNCHRONIZATION

PRIORITY:

AIRFLOW > FUEL

NEVER:

- excessive IQ before airflow
- abrupt torque spike
- unstable vane transition
- smoke-rich torque request

AUTOMATIC TRANSMISSION RULES

- smooth torque rise
- stable converter loading
- reduced harsh shifts
- controlled transient torque

MANUAL TRANSMISSION RULES

- smoother pull behavior
- stable transient response
- cleaner gear transition
- smoother midrange torque

SMOKE REDUCTION STRATEGY

BLACK SMOKE:

- reduce delta aggression
- smooth transient fuel
- preserve airflow progression

WHITE SMOKE:

- stabilize startup fueling
- smooth low-temp transitions
- reduce harsh enrichment

FUEL KNOCK REDUCTION

- smooth delta behavior
- stabilize low-load progression
- reduce transient spikes

RATTLE REDUCTION STRATEGY

TARGET:

- cold start
- idle
- low-load pull
- 1000–2000 RPM

FIX:

- smoother torque progression
- stabilized combustion pressure rise

BLAKKBOX FILTERING ENGINE

Core functions:

- combustion-safe refinement
- spike suppression
- continuity stabilization
- interpolation-safe reconstruction
- torque smoothing
- smoke suppression

HEX INTELLIGENCE ENGINE

- modified region analysis
- entropy scoring
- binary heatmaps

- probable axis detection
- structure warning detection

CONFIDENCE SCORING SYSTEM

Scoring factors:

- pointer validity
- axis monotonicity
- smoothness
- alignment correctness
- XREF support
- statistical consistency
- checksum compatibility

CONFIDENCE LEVELS

90–100:

CONFIRMED

75–89:

HIGHLY PROBABLE

50–74:

PROBABLE

<50:

WEAK CANDIDATE

DELTA STATISTICS ENGINE

Generate:

- modified byte count
- grouped surface count
- continuity metrics
- smoothing statistics
- OEM preservation ratio

REGRESSION VALIDATION

Verify:

- stable parser behavior
- stable grouping behavior
- stable interpolation continuity
- no geometry corruption
- no offset drift

STRUCTURED EXPORTS

Generate:

- region map
- axis definitions
- calibration candidates
- integrity summaries
- confidence scores
- pointer graphs

EXPORT FORMATS

- JSON
- CSV
- XML
- SQLite

FINAL EXPORT RULES

EXPORT:

- FINAL BIN
- CSV REPORT
- PDF WORKFLOW

CHECKSUM:

DO NOT CORRECT DURING PROCESSING

FINAL CHECKSUM:

MANUAL BITEDIT CHECKSUM ONLY

FINAL VALIDATION CHECKLIST

VERIFY:

- ROM size correct
- SW ID correct
- no offset drift
- no corrupted regions
- no modified zero-groups
- interpolation preserved
- no spikes
- no harsh transitions
- smoke reduction applied
- combustion refinement applied
- checksum untouched

FINAL PHILOSOPHY

GOOD DENSO CALIBRATION IS:

- combustion efficient
- smoke controlled
- torque stable
- OEM structured
- interpolation clean
- thermally controlled
- mechanically sympathetic
- stable during low-load operation
- safe during aggressive pull

BAD CALIBRATION IS:

- smoke rich
- torque spiky
- rail aggressive
- unstable transient
- noisy combustion
- broken interpolation
- harsh low-load behavior
- unstable startup behavior